- Amazon CloudWatch
- Auto Scaling groups
- Amazon Simple Queue Service (SQS)
- AWS CloudFormation
- AWS Identity and Access Management (IAM, not bound to a single region; if you create an IAM user, that user is available in all regions)

Chapter requirements

To fully understand this chapter, you need to have read and understood the following concepts:

- EC2 (chapter 3)
- Autoscaling (chapter 13)
- Elastic Load Balancing (chapter 14)
- Simple Queue Service (chapter 14)

On top of that, the example included in this chapter makes intensive use of the following:

- DynamoDB (chapter 12)
- Express, a Node.js web application framework

In this chapter, you'll learn everything you need to design a fault-tolerant web application based on EC2 instances (which aren't fault tolerant by default). During this chapter, you will build a fault-tolerant web application that allows a user to upload an image, apply a sepia filter to the image, and download the image. First, you will learn how to distribute a workload among multiple EC2 instances. Instead of